HYDRAULIC VARIABLE VALVE TIMING CONTROL [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

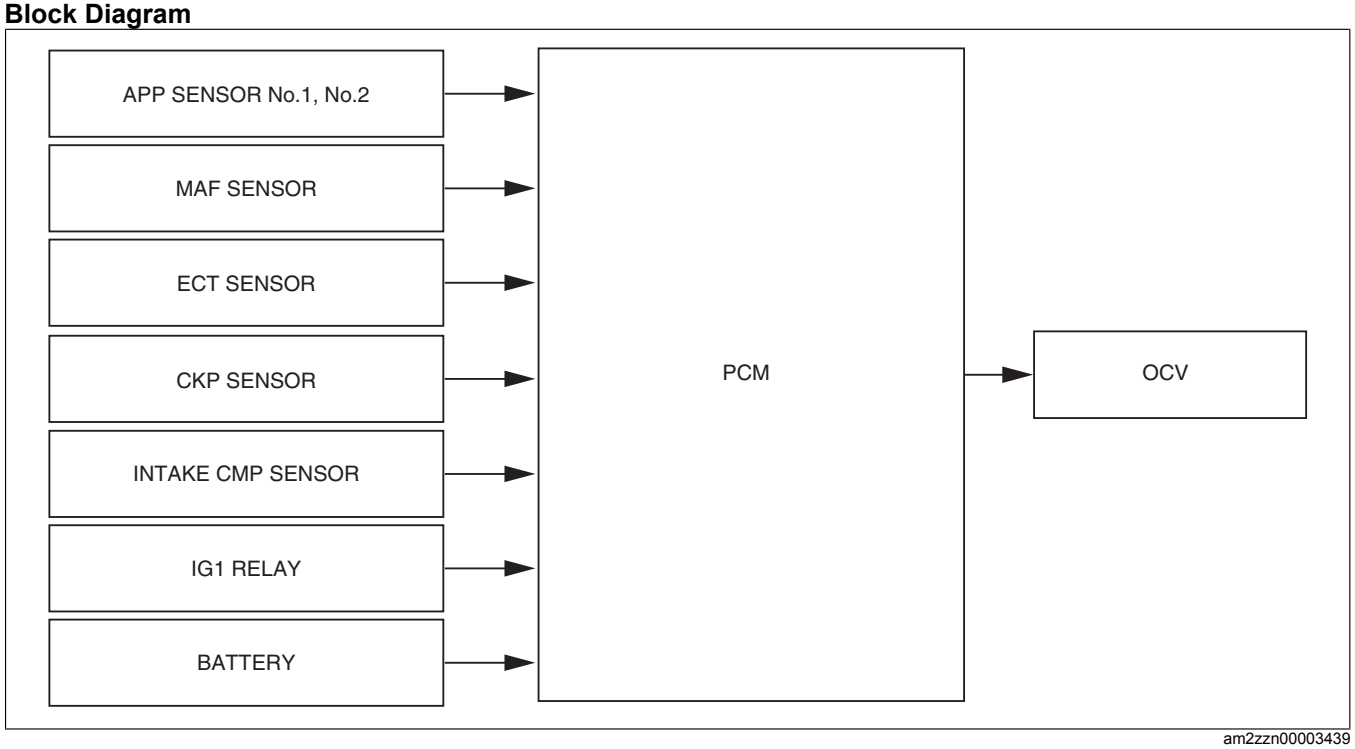
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SKYACTIV-G 1.3, SKYACTIV-G 1.5 (With 4-1 Exhaust System)

Outline

- Changes the intake valve timing according to engine operation conditions to improve engine output, fuel economy, and emission performance.
- Based on each input signal, the PCM determines the optimum intake valve timing according to the engine operation conditions. The PCM drives the OCV, and switches the oil passages of the hydraulic variable valve timing actuator to control the intake valve timing at optimum.

Block Diagram



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Operation

Intake valve timing determination

- The PCM controls current to the OCV to obtain optimum intake valve timing according to the engine operation conditions (target intake valve timing).
- In addition to controlling the OCV drive current value to obtain the target intake valve timing, the PCM compares the value with the actual intake valve timing to correct the OCV drive current value if necessary.

Target intake valve timing

- The target intake valve timing is determined by the engine speed and the charging efficiency.

Actual intake valve timing

- The actual intake valve timing is calculated by subtracting the cam maximum retard learning value from the value (advance amount) detected by the intake CMP sensor and CKP sensor.

Cam maximum retard learning value

- Though the intake camshaft valve timing (including maximum retard position) is detected based on the difference in the rise between the intake CMP sensor and the CKP sensor signals, variation in signals from the assembly of each sensor occurs. As a result, the PCM stores the difference in the rise of the intake CMP sensor and crankshaft position signals to prevent detection of variations in the intake valve timing.

OCV drive current determination

- The PCM divides the oil control valve OCV drive range into three modes according to engine operation conditions. The OCV drive current is determined based on the target current calculated in each mode.

Range mode table

Mode name	Control description
Feedback mode	• The feedback mode constantly monitors whether the target intake valve timing is the determined value according to the engine operation conditions and controls the OCV drive current based on the results. **Mode execution condition** — Except cleaning mode, cam maximum retard mode

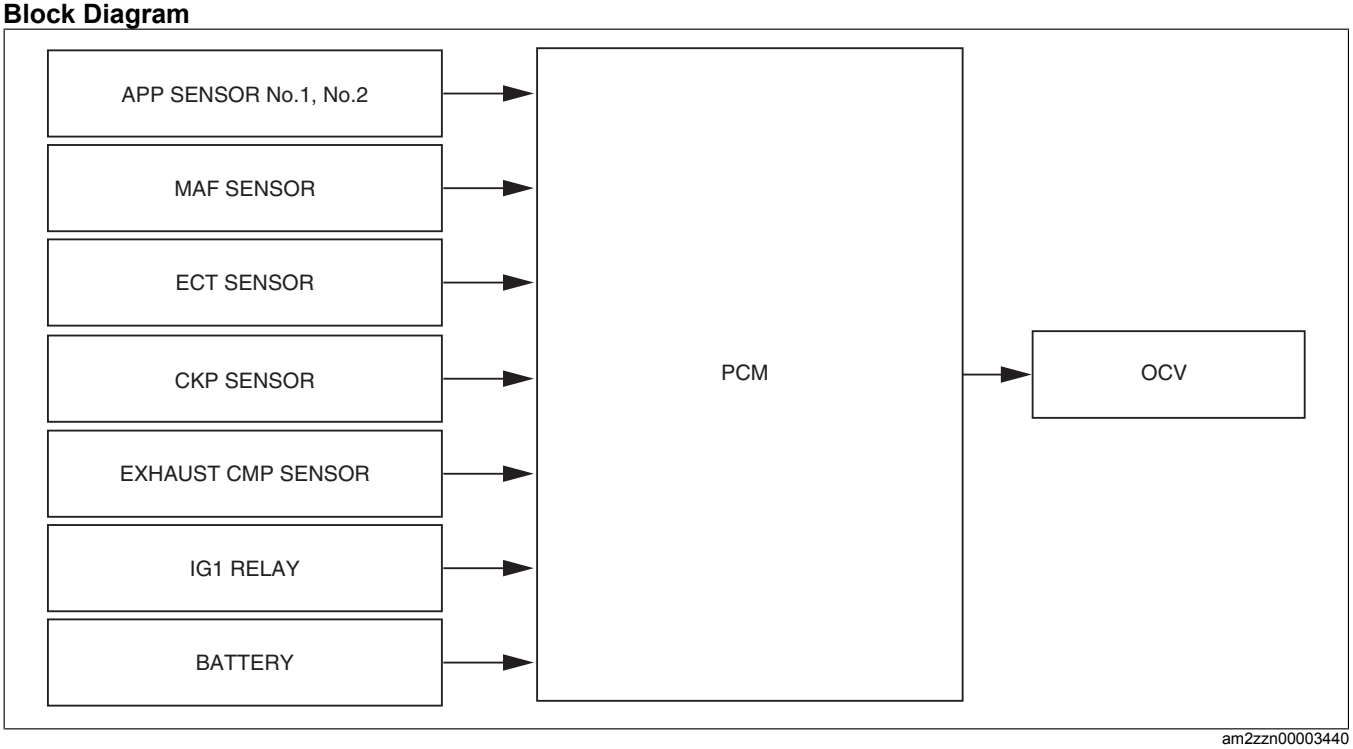
Mode name	Control description
Cleaning mode	• Cleaning mode is to remove foreign matter in the OCV oil passages. **Mode execution condition** — During deceleration fuel cut — Engine speed: **2,250—3,500 rpm** — Engine coolant temperature: **80 °C {176 °F} or more**
Cam retard mode	• If the cam retard is appropriate according to engine operation conditions such as while in torque down execution during idling, the cam retard mode retards the intake valve timing and stabilizes the engine speed. **Mode execution condition** • When any of the following conditions is met: — During engine cranking — Engine speed: **Less than 750 rpm** — Engine coolant temperature: **Less than 20 °C {68 °F}** — Other conditions such as when maximum retard is appropriate according to engine rotation and load conditions

SKYACTIV-G 1.5 (With 4-2-1 Exhaust System)

Outline

- Changes the exhaust valve timing according to engine operation conditions to improve engine output, fuel economy, and emission performance.
- Based on each input signal, the PCM determines the optimum exhaust valve timing according to the engine operation conditions. The PCM drives the OCV, and switches the oil passages of the hydraulic variable valve timing actuator to control the exhaust valve timing at optimum.
- Based on the coordination with the electric variable valve timing control and by increasing the amount of overlap during high engine loads, nitrogen oxide (NOx) occurring largely at high temperatures is reduced by re-circulating exhaust gas into the combustion chamber, which also reduces combustion temperature.

Block Diagram



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Operation

Exhaust valve timing determination

- The PCM controls current to the OCV to obtain optimum exhaust valve timing according to the engine operation conditions (target exhaust valve timing).
- In addition to controlling the OCV drive current value to obtain the target exhaust valve timing, the PCM compares the value with the actual exhaust valve timing to correct the OCV drive current value if necessary.

Target exhaust valve timing

- The target exhaust valve timing is determined by the engine speed and the charging efficiency.

Actual exhaust valve timing

- The actual exhaust valve timing is calculated by subtracting the cam maximum advance learning value from the value (retard amount) detected by the exhaust CMP sensor and CKP sensor.

Cam maximum advance learning value

- Though the exhaust camshaft valve timing (including maximum advance position) is detected based on the difference in the rise between the exhaust CMP sensor and the CKP sensor signals, variation in signals from the assembly of each sensor occurs. As a result, the PCM stores the difference in the rise of the exhaust CMP sensor and crankshaft position signals to prevent detection of variations in the exhaust valve timing.

OCV drive current determination

- The PCM divides the oil control valve OCV drive range into three modes according to engine operation conditions. The OCV drive current is determined based on the target current calculated in each mode.

Range mode table

Mode name	Control description
Feedback mode	• The feedback mode constantly monitors whether the target exhaust valve timing is the determined value according to the engine operation conditions and controls the OCV drive current based on the results. **Mode execution condition** — Except cleaning mode, cam maximum advance mode
Cleaning mode	• Cleaning mode is to remove foreign matter in the OCV oil passages. **Mode execution condition** — During deceleration fuel cut — Engine speed: **2,250—3,500 rpm** — Engine coolant temperature: **80 °C {176 °F} or more**
Cam advance mode	• If the cam advance is appropriate according to engine operation conditions such as while in torque down execution during idling, the cam advance mode advances the exhaust valve timing and stabilizes the engine speed. **Mode execution condition** • When any of the following conditions is met: — During engine cranking — Engine speed: **Less than 875 rpm** — Engine coolant temperature: **Less than 35 °C {95 °F}** — Other conditions such as when maximum advance is appropriate according to engine rotation and load conditions